

Scheme of Valuation/Answer Key													
(Scheme of evaluation (marks in brackets) and answers of problems/key)													
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY													
FIFTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2022													
Course Code: EET303													
Course Name: MICROPROCESSOR AND MICROCONTROLLERS													
Max. Marks: 100			Duration: 3 Hours										
PART A													
		Answer all questions, each carries 3 marks.	Marks										
1		Describe multiplexing and de multiplexing process in 8085 μP Explanation of Multiplexing and de multiplexing – 1.5 Mark each A multiplexer is a system of multiple inputs and just one output to receive signals coming from multiple acquisition networks. The device transfers all input signals to a microprocessor, which receives and processes the data, transmits it to the output devices, and controls the system as a whole. Multiplexing is a process in which various data signals are integrated to give a single output. De multiplexing is a process in which one input data signal is divided into various output signals. It takes various input signals and gives one output signal only.	(3)										
2		Write any three addressing modes of 8085 μP with examples. Any three addressing modes with examples, 1mark for each Immediate Addressing Mode - In this type of addressing mode the operand is specified in the instruction itself. The data will be part of the program instruction - Consider this instruction: ADI 34H – This instruction adds the immediate data, 34H to the accumulator. Direct Addressing Mode - In this mode of addressing, the address of the data (operand) is specified within the instruction. Consider the instruction STA 2000H Register Addressing Mode - In this mode, the data is copied from one register to another. For example: MOV A, B; means data in register B is copied to register A.	(3)										
3		Write an 8085 ALP to find 2 s complement of an 8-bit number which is stored in location 2050H. <table><tr><td>LDA 2500</td><td>Load the value to Acc.</td></tr><tr><td>CMA</td><td>Complement Accumulator contents.</td></tr><tr><td>INR A</td><td>Increment value in Accumulator.</td></tr><tr><td>STA 2050</td><td>Store the value of A-reg to memory address</td></tr><tr><td>HLT</td><td></td></tr></table>	LDA 2500	Load the value to Acc.	CMA	Complement Accumulator contents.	INR A	Increment value in Accumulator.	STA 2050	Store the value of A-reg to memory address	HLT		(3)
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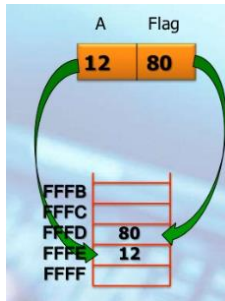
4	What is stack? How stack is initialised? The stack is a LIFO (last in, first out) data structure implemented in the RAM area and is used to store addresses and data when the microprocessor branches to a subroutine. Then the return address used to get pushed on this stack. Also to swap values of two registers and register pairs we use the stack as well.	(3)								
5	Distinguish between vectored and non-vectored interrupts Vectored Interrupts A vectored interrupt is where the CPU actually knows the address of the Interrupt Service Routine in advance. All it needs is that the interrupting device sends its unique vector via a data bus and through its I/O interface to the CPU. The CPU takes this vector, checks an interrupt table in memory, and then carries out the correct ISR for that device. So the vectored interrupt allows the CPU to be able to know what ISR to carry out in software (memory). Non Vectored Interrupts A non-vectored interrupt is where the interrupting device never sends an interrupt vector. An interrupt is received by the CPU, and it jumps the program counter to a fixed address in hardware. This is literally a hard coded ISR which is device agnostic. MIPS uses this via it's syscall instruction, which is the same instruction you use regardless what device, if it is an external event notifier (e.g. keyboard input), a completion signal (e.g. print completion), a clock interrupt to tell the CPU to allocate control to a different device, or an abnormal event indicator (e.g. power failure). The CPU crucially does not know which device caused the interrupt without polling each I/O interface in a loop and checking the status register of each I/O interface to find the one with status "interrupt created".	(3)								
6	Write the differences between microprocessor and microcontroller. <table><thead><tr><th>Microprocessor</th><th>Microcontroller</th></tr></thead><tbody><tr><td>Microprocessors can be understood as the heart of a computer system.</td><td>Microcontrollers can be understood as the heart of an embedded system.</td></tr><tr><td>A microprocessor is a processor where the memory and I/O component are connected externally.</td><td>A microcontroller is a controlling device wherein the memory and I/O output component are present internally.</td></tr><tr><td>The circuit is complex due to external connection.</td><td>Microcontrollers are present on chip memory. The circuit is less complex.</td></tr></tbody></table>	Microprocessor	Microcontroller	Microprocessors can be understood as the heart of a computer system.	Microcontrollers can be understood as the heart of an embedded system.	A microprocessor is a processor where the memory and I/O component are connected externally.	A microcontroller is a controlling device wherein the memory and I/O output component are present internally.	The circuit is complex due to external connection.	Microcontrollers are present on chip memory. The circuit is less complex.	(5)
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7	Explain the indexed addressing modes of 8051 μC with two examples - Only program memory can be accessed in the index addressing. Either the DPTR or PC can be used as an index register. The destination operand is always the register A. Example - MOV A, @ A+DPTR - This instruction copies the contents of memory locations pointed by sum of	(5)								

		the accumulator A and the register DPTR into the accumulator A.									
8		Differentiate between CLR C and CPL C instruction of 8051 μC CLR C The CLR instruction sets the specified destination operand to a value of 0. CPL C The CPL instruction logically complements the value of the specified destination operand and stores the result back in the destination operand. Bits that previously contained a 1 will be changed to a 0 and bits that previously contained a 0 will be changed to a 1.	(3)								
9		Explain SCON register in 8051 μC - Serial port control and status register is the special function register SCON. - This register contain not only the mode selection bits but also the 9th data bit for transmit and receive (TB8 and RB8) and the serial part interrupt bits (TI and RI) (MSB) <table border="1"><tr><td>SM0</td><td>SM1</td><td>SM2</td><td>REN</td><td>TB8</td><td>RB8</td><td>TI</td><td>RI</td></tr></table> (LSB) SCON (Serial Port Control Register) SM0 – Serial port mode 0 shift register SM1 – serial port mode 1 8 bit UAR + variable SM2 – enable multiprocessor communication in mode 2/3 REN – set/ clear by software to enable/disable reception TB8 – the 9th bit that will be transmitted in mode 2/3 set/clear by software. RB8 – in mode 2/3 it is the 9th bit that was received in mode 1 if SM2 =0, RB8 is the stop bit that was received in mode it is not used. TI – transmit interrupt flag set by hardware at the end of 8 bit time in mode 0 at the beginning of the stop bit in the other mode it must be cleared by software RI – receive interrupt flag set by hardware and must be cleared by software.	SM0	SM1	SM2	REN	TB8	RB8	TI	RI	(3)
SM0	SM1	SM2	REN	TB8	RB8	TI	RI				
10		Indicate which mode and which timers are selected for each of the following. (a) MOV TMOD, #01H (b) MOV TMOD, #20H (a) TMOD = 00000001, mode 1 of timer 0 – 1.5 marks (b) TMOD = 00100000, mode 2 of timer 1 – 1.5 marks	(3)								
PART B											
<i>Answer any two full questions, each carries10 marks.</i>											
11	a)	Point out the importance of the following functional blocks of 8085Mp architecture (i) Flag flip flops (ii) Instruction decoder and machine cycle encoding	(8)								

	(iii) Stack pointer (iv) Program counter (i) Flag Register <table><tr><th>D7</th><th>D6</th><th>D5</th><th>D4</th><th>D3</th><th>D2</th><th>D1</th><th>D0</th></tr><tr><td>S</td><td>Z</td><td></td><td>AC</td><td></td><td>P</td><td></td><td>CY</td></tr></table> - S- Sign flag After the execution of arithmetic or logical operations, if D7 of the result is 1, the sign flag is set. In a given byte if D7 is 1, the number will be viewed as negative number. If D7 is 0, the number will be considered as positive number. - Zero flag (Z) If the result stored in an accumulator is zero then this flip flop is set otherwise it is reset. - Auxiliary Carry(AC) If any carry goes from D3 to D4 in the output then it is set otherwise it is reset. - Parity flag If the no of 1's even in the output stored in the accumulator then it is set otherwise it is reset for the odd. - Carry flag If the result stored in an accumulator generates a carry from D7 in its final output then it is set. otherwise it is reset. (ii) Instruction decoder and machine cycle encoding It is an 8 bit register. When an instruction is fetched from memory, then it is stored in the Instruction register. Instruction decoder decodes the information present in the Instruction register. The instruction register not programmable and cannot be accessed through any instruction This accepts a bit pattern (OPCODE) from instruction register, decodes it and gives the decoded information to control logic. The information includes what operation is to be performed, who is going to perform it, how many operand bytes the instruction contains, etc. (iii) Stack pointer - Stack pointer It is a 16 bit registers which acts like a stack i.e., this is used to arrange the data neatly. It is incremented or decremented by 2 during PUSH and POP operation (iv) Program counter It is a 16 bit registers, which is used to store the memory address location of the next instruction to be executed	D7	D6	D5	D4	D3	D2	D1	D0	S	Z		AC		P		CY	
D7	D6	D5	D4	D3	D2	D1	D0											
S	Z		AC		P		CY											
b)	Explain the basic steps involved in reading from and writing to memory locations There are two special registers, the memory address register (MAR) and the memory data register (MDR), and a signal indicating whether to read from the memory or write to the memory. The memory itself consists of a number $n =$	(6)																

		2A of memory registers, each of which holds B bits. Read–write memory, or RWM is a type of computer memory that can be easily written to as well as read from using electrical signaling normally associated with running a software, and without any other physical processes.										
12	a)	Draw and explain the timing diagram of OUT 00H instruction of 8085Mp Timing Diagram of OUT instruction in 8085 ❖ OUT (8 bit Port Address = 05H) [Opcode = D3H] PC = 1000H 8085 A = 24H <table><thead><tr><th></th><th>Add</th><th>Data</th></tr></thead><tbody><tr><td></td><td>1000H</td><td>D3H</td></tr><tr><td></td><td>1001H</td><td>05H</td></tr></tbody></table> ❖ OUT 05H executions needs opcode fetch, Memory Read & IO Write machine cycle. ❖ So, After instruction, A = 24H will get display at 05H port Address. ❖ OUT 05H [Opcode Fetch = D3H] ❖ Memory Read 05H from 1001H ❖ IO Write A at 05H Output Port The diagram shows the following sequence of events: T1: ALE is active high. A15-A8 = 10H, AD7-AD0 = 00H. $IO/\overline{M}$ is active low, $S1=1, S0=1$. T2: A15-A8 = 10H, AD7-AD0 = D3H (Opcode). RD is active low. T3: A15-A8 = 10H, AD7-AD0 = D3H (Opcode). RD is active low. T4: A15-A8 = 10H, AD7-AD0 = 01H. RD is active low. T5: A15-A8 = 10H, AD7-AD0 = 05H. RD is active low. T6: A15-A8 = 10H, AD7-AD0 = 05H. RD is active low. T7: A15-A8 = 10H, AD7-AD0 = 05H. RD is active low. T8: A15-A8 = 10H, AD7-AD0 = 05H. RD is active low. T9: A15-A8 = 10H, AD7-AD0 = 05H. RD is active low. T10: A15-A8 = 10H, AD7-AD0 = 05H. RD is active low. WR becomes active high. $IO/\overline{M}$ is active low, $S1=0, S0=1$.		Add	Data		1000H	D3H		1001H	05H	(8)
	Add	Data										
	1000H	D3H										
	1001H	05H										
	b)	Discuss instruction cycle, machine cycle and T state of 8085 Instruction Cycle - An instruction cycle can be defined as total time required by the microprocessor to completely fetch and execute an instruction - An instruction cycle is comprised of one or more number of machine cycle Machine Cycle - It is the time required by the microprocessor to complete the operation of accessing the memory devices or I/O devices. T State One complete cycle of clock is called as T-state	(6)									
13	a)	Two 16-bit numbers are stored in memory location 8000H onwards. Write an 8085 ALP to interchange the 16-bit numbers stored in the memory <pre>LHLD 8000H XCHG LHLD 8002H SHLD 8000H XCHG SHLD 8002H HLT</pre>	(7)									
14	a)	List the sequence of operations involved during the execution of PUSH PSW and POP PSW instructions of 8085 Mp	(8)									

PUSH PSW



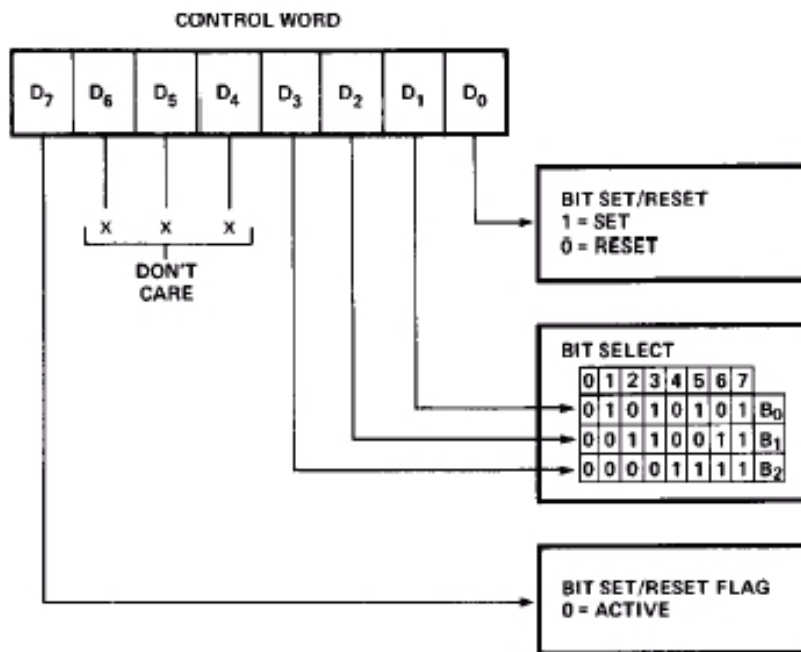
PUSH and POP. In case of PUSH operation, the SP register gets decreased by 2 and new data item used to insert on to the top of the stack. On the other hand, in case of POP operation, the data item will have to be deleted from the top of the stack and the SP register will get increased by the value of 2.

- b) Write an 8085 ALP to save the content of accumulator and flag register to BC register pair after adding 25H and 65H. (6)

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LXI H 4000H : "HL points 4000H"
MOV A, M    : "Get first operand"
INX H       : "HL points 4001H"
ADD M       : "Add second operand"
STAX B      : "Store result at 4002H"
HLT         : "Terminate program execution"
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- 15 a) Explain all modes of operation of 8255 PPI

BSR Mode



(8)

		I/O Mode CONTROL WORD <table><tr><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td></tr></table> GROUP B PORT C (LOWER) 1 = INPUT 0 = OUTPUT PORT B 1 = INPUT 0 = OUTPUT MODE SELECTION 0 = MODE 0 1 = MODE 1 GROUP A PORT C (UPPER) 1 = INPUT 0 = OUTPUT PORT A 1 = INPUT 0 = OUTPUT MODE SELECTION 00 = MODE 0 01 = MODE 1 1X = MODE 2 MODE SET FLAG 1 = ACTIVE	D7	D6	D5	D4	D3	D2	D1	D0	
D7	D6	D5	D4	D3	D2	D1	D0				
16	a)	List out the applications and challenges in Embedded Systems. - The embedded systems have a huge variety of application domains. So, the embedded systems span all aspects of our modern life. The following are various applications of embedded systems. Home Appliances Dishwasher, washing machine, microwave, Top-set box, security system, DVD, garden sprinkler systems etc.. Office Automation Fax, copy machine, smart phone system, modern, scanner, printers. Security Face recognition, finger recognition, eye recognition, building security system, airport security system, alarm system. Academia Smart board, smart room, calculator, smart cord. Instrumentation Signal generator, signal processor, power supplier, Process instrumentation, Telecommunication Router, hub, cellular phone, IP phone, web camera Automobile: Fuel injection controller, anti-locking brake system, airbag system, GPS, cruise control. Entertainment: MP3, video game, smart toy. Aerospace: Navigation system, automatic landing system, flight attitude controller, space explorer, space robotics. Industrial automation: Assembly line, data collection system, monitoring systems on pressure, voltage, current, temperature, hazard detecting system, industrial robot. Personal: PDA, iPhone, palmtop, data organizer.	(6)								
16	b)	Draw and explain memory organisation in 8051 μ C									

	The diagram illustrates the memory organization of the 8051 microcontroller. It is divided into two main 128B sections: Lower 128B (00H - 7FH) and Upper 128B (80H - FFH). The Lower 128B contains 80 General Purpose Registers (00H-7FH), 16 Bit-Addressable Registers (20H-2FH), and four Register Banks (00H-0FH). The Upper 128B contains Special Function Registers (SFRs) and 128B Additional Memory. The SFRs are located at 80H-FFH and include ACC, PSW, IP, P3, IE, P2, SBUF, SCON, P1, TH1, TH0, TL1, TL0, TMOD, TCON, PCON, DPH, DPL, SP, and P0. Lower 128B (00H - 7FH) Upper 128B (80H - FFH) (Direct and Indirect Addressing) (Direct Addressing) (Indirect Addressing) - The 8051 has 4K bytes of internal program memory(ROM) and 256 bytes of internal data memory(RAM). - The Program Memory of the 8051 Microcontroller is used for storing the program to be executed i.e. instructions. The Data Memory on the other hand, is used for storing temporary variable data and intermediate results. Data Memory (RAM) of 8051 Microcontroller - The 8051's on-chip memory consists of 256 memory bytes organized as follows: First 128 bytes: 00h to 1Fh Register Banks 20h to 2Fh Bit Addressable RAM 30 to 7Fh General Purpose RAM Next 128 bytes: 80h to FFh Special Function Registers	
Module 4		
<i>Answer anyone full questions, each carries 14 marks.</i>		
17	a) Write an 8051 ALP to multiply A1H and B1H and store the LSB and MSB results in R0 and R1 registers of bank 2. <pre> MOV R0, #A1H ; Get first no. in R0 MOV A, R0 ; Copy R0 value to accumulator MOV R1, #B1H ; get second no in R1 MOV B, R1 ; Copy R1 value to B register MUL AB ; Multiply A by B MOV 60H, A ; Output stored in 60H data location </pre>	(8)
	b) Explain the operation of following instructions of 8051 μC (a) DA A (b) SUBB A, @R0 (c) ANL A, #10H (a) DA A - Decimal Adjust Accumulator (b) SUBB A, @ R0	(3)

		Subtracts the contents of memory location whose address is given by R0 and carry together from A and stores result in A (c) ANL A, #10H Logically ANDs contents of A with 10H and stores result in A	
18	a)	Explain I/O ports and its functions in 8051 μC • There are totally four 8-bit I/O Ports namely PORT 0 , PORT 1 , PORT 2 , PORT 3. • Three out of four ports namely PORT 0 , 2 and 3 are multifunctional PORTS. • Except port 1 all other ports have alternate functions • Port 1 can be used only for IO Operation. • When external memory is employed, port 0 functions as multiplexed low byte address or data lines and port 2 functions as high byte address lines. • Port 3 serves for (i) Serial data communication (ii) External memory Read and Write control signals • Also the port pin P3.7 functions as read control and the port pin P3.6 functions as write control for data memory.	(6)
	b)	Write an 8051 C program to get a byte of data from P0. If it is less than 100, send it to P1; otherwise, send it to P2. <pre>#include <reg51.h> Void main(void) { unsigned char mybyte; P0 = 0xFF; While (1) { mybyte=P0; If (mybyte<100) P1 = mybyte; Else P2=mybyte; } }</pre>	(4)
Module 5			
<i>Answer any two full questions, each carries 10 marks.</i>			
19	a)	Illustrates different modes of operation of timers in 8051 μC • Both timers 0 and 1 use the same register, called TMOD, to set the various timer operation modes. • TMOD is an 8 bit register in which the lower 4 bits are set aside for Timer 0 and the upper 4 bits for Timer 1. • In each case, the lower 2 bits are used to set the timer mode and the upper 2 bits to specify the operation. M1, M0 • M0 and M1 select the timer mode. As shown in figure, there are three modes: 0, 1 and 2. • Mode 0 is a 13 bit timer, mode 1 is a 16 bit timer, and mode 2 is an 8 bit	(5)

		timer. • TMOD is an 8-bit register used for selecting timer or counter and mode of timers. • Lower 4-bits are used for control operation of timer 0 or counter0, and remaining 4-bits are used for control operation of timer1 or counter1. • This register is present in SFR register, the address for SFR register is 89th.	
16	a)	Draw and explain interfacing of DAC with 8051 μC • Ideally we connect the output pin I_{out} to a resistor, convert this current to voltage, and monitor the output on the scope. • In real life, however, this can cause inaccuracy since the input resistance of the load where it is connected will also affect the output voltage. • For this reason, the I_{ref} current output is isolated by connecting it to an op amp such as the 741 with $R_f = 5K$ ohms for the feedback resistor. • For generating sinewave, at first we need a look-up table to represent the magnitude of the sine value of angles between 0° to 360°. The sine function varies from -1 to +1. • In the table only integer values are applicable for DAC input. In this example we will consider 30° increments and calculate the values from degree to DAC input. • We are assuming full-scale voltage of 10V for DAC output. We can follow this formula to get the voltage ranges.	(6)
20	b)	List the interrupts in 8051	(3)
	(i)	Interrupts are basically the events that temporarily suspend the main program, pass the control to the external sources and execute their task. It then passes the control to the main program where it had left off. 8051 has five interrupts. These interrupts are INT0, INT1, TO, T1, TI/RI.	
	(ii)	Write the steps in execution of an interrupt in 8051 1. It finishes the instruction it is executing and saves the address of the next instruction (PC) on the stack. 2. It also saves the current status of all the interrupts internally (i.e., not on the stack).	(5)

	3. It jumps to a fixed location in memory called the interrupt vector table that holds the address of the interrupt service routine. 4. The microcontroller gets the address of the ISR from the interrupt vector table and jumps to it. It starts to execute the interrupt service subroutine until it reaches the last instruction of the subroutine, which is RETI (return from interrupt). Upon executing the RETI instruction, the microcontroller returns to the place where it was interrupted. First, it gets the program counter (PC) address from the stack by popping the top two bytes of the stack into the PC. Then it starts to execute from that address.	
